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$7x$ is an algebraic term in which 7 is a _____ and x is a _____.

Answer (Please select your correct option)



term, expression



coefficient, variable



variable, coefficient



numerical, alphabet

Correct Answer Solved By Hadi
usmanraj20@gmail.com
03228043306

Made By: Waqar Siddhu



Which of the following is the pivot element in the second row of the matrix?

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 8 \\ 0 & 0 & -3 & 5 & 1 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & -3 \end{pmatrix}$$

Answer (Please select your correct option)

0



1



-3



5



Correct Answer Solved By Hadi
usmanraj20@gmail.com
03228043306

Made By: Waqar Siddhu



Which of the following is true for the matrix $\begin{pmatrix} 1 & 3 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$?

Answer (Please select your correct option)

☐ It is an identity matrix.

☐ It is in reduced echelon form.

☐ It is in echelon form.

☐ It is a rectangular matrix.

Correct Answer Solved By Hadi
usmanraj20@gmail.com
03228043306

Made By: Waqar Siddhu



If reduced echelon form of a linear system is $\begin{bmatrix} 1 & 0 & 5 & 5 \\ 0 & 1 & 1 & 6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ when free variable $x_3 = 0$, then which of the following is true for it?

Answer (Please select your correct option)

☐ The particular solution is $(0, 5, 6)$.

☐ The particular solution is $(6, 5, 0)$.

☐ The particular solution is $(5, 6, 0)$.

☐ The particular solution is $(0, 6, 5)$.

Correct Answer Solved By Hadi
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If $\vec{b} = c_1 \vec{u}_1 + c_2 \vec{u}_2$ and $\vec{a} = d_1 \vec{u}_1 + d_2 \vec{u}_2$ where c_1, c_2, d_1 and d_2 are scalars, then which of the following options is correct?

Answer (Please select your correct option)

☐ Only $\vec{b} \in \text{Span}\{\vec{u}_1, \vec{u}_2\}$.

☐ Only $\vec{a} \in \text{Span}\{\vec{u}_1, \vec{u}_2\}$.

☐ Both $\vec{a}, \vec{b} \in \text{Span}\{\vec{u}_1, \vec{u}_2\}$.

☐ Both $\vec{a}, \vec{b} \notin \text{Span}\{\vec{u}_1, \vec{u}_2\}$.

Correct Answer Solved By Hadi
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03228043306

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If a homogeneous system $Ax = 0$ has a trivial solution, then which of the following is (are) the value(s) of the vector x ?

Answer (Please select your correct option)

☐ -1

☐ 0

☐ 1

☐ 2

Correct Answer Solved By Hadi
usmanraj20@gmail.com
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If a homogeneous system $AX = 0$ has non-trivial solution, then which of the following is true for the system?

Answer (Please select your correct option)



The system has at least no free variable.



The system has at least one free variable.

Correct Answer Solved By Hadi
usmanraj20@gmail.com
03228043306



The system has at least two free variables.



The system has at least three free variables.

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If $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1+6x_3 \\ 3 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 6 \\ 0 \\ 1 \end{bmatrix}$ be the general solution of $Ax = b$, then which of the following is parametric equation for the given solution?

Answer (Please select your correct option)

☐

$$p = s x + t u + v$$

☐

$$x = p + t v$$

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☐

$$v = p + t x$$

☐

$$t x = p + v$$



If the equation $T(\vec{x}) = \vec{0}$ has only the trivial solution, then which of the following is true for the linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$?

Answer (Please select your correct option)



T is one-to-one.

Correct Answer Solved By Hadi
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T is onto.



T is a rotation.



T is a reflection.

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If linear transformation $T : R^n \rightarrow R^m$ is transformed into a matrix A , then which of the following is the order of A ?

Answer (Please select your correct option)

☐ $n \times m$

☐ $m \times m$

☐ $m \times n$

☐ $n \times n$

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If $A = A^t$ (where A is a square matrix) , then which of the following is the most appropriate option for A ?

Answer (Please select your correct option)



A is an invertible matrix.



A is a singular matrix.



A is a symmetric matrix.

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A is a scalar matrix.

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If A is an invertible matrix , then which of the following is true ?

Answer (Please select your correct option)

☐ $(A^{-1})^{-1} = A$

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☐ $(A^{-1})^{-1} = \frac{1}{A}$

☐ $(A^{-1})^{-1} = A^{-1}$

☐ $(A^{-1})^{-1} = \det(A)$



If $A = \begin{bmatrix} 4 & -1 \\ 5 & 3 \end{bmatrix}$, then which of the following is the value of $\det(A)$?

Answer (Please select your correct option)

☐

7

☐

-17

☐

17

☐

11

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If a system of equations is solved using the Jacobi's method , then which of the following is NOT true about the matrix M that is derived from the coefficient matrix ?

Answer (Please select your correct option)



All of its entries below the diagonal must be zero .



All of its entries above the diagonal must be zero .



It may or may not be invertible .



It is a non-singular matrix .

Correct Answer Solved By Hadi
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If the matrix $A = \begin{bmatrix} 5 & 4 \\ 3 & 3 \end{bmatrix}$, then which of the following is the most suitable option for it ?

Answer (Please select your correct option)

☐

It is not strictly diagonally dominant.

Correct Answer Solved By Hadi
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☐

It is strictly diagonally dominant.

☐

It is not diagonally dominant.

☐

It is diagonally dominant.

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If $A = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$, then which of the following is the minor of entry a_{21} ?

Answer (Please select your correct option)

2



3



4



5



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If $A = \begin{bmatrix} 4 & 5 \\ 6 & 7 \end{bmatrix}$, then which of the following is the value of C_{21} ?

Answer (Please select your correct option)



5



-5



6



-6

Correct Answer Solved By Hadi
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If the determinant of the matrix $A = \begin{bmatrix} 2 & 3 & 4 \\ 1 & 2 & 2 \\ 3 & 4 & 5 \end{bmatrix}$ is -1 and the matrix B is obtained by adding 2 times of the second row in the first row of the matrix A , then which of the following is true about the matrix B ?

Answer (Please select your correct option)

- ☐ Its determinant is -1 .
- ☐ Its determinant is 1 .
- ☐ Its determinant can not be evaluated .
- ☐ The information is not sufficient to calculate the determinant .

Correct Answer Solved By Hadi
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03228043306



If the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix}$, then which of the following is true about it ?

Answer (Please select your correct option)



Its determinant is 0 .



Its determinant is 1 .



Its determinant is 2 .

Correct Answer Solved By Hadi
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Its determinant is 4 .

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If a set $S = \{ v_1 = (1, 2), v_2 = (4, 8) \}$, then which of the following is the most appropriate option ?

Answer (Please select your correct option)



It is a basis of R^2 .



It is linearly independent.



It spans R^2 .



It is linearly dependent.

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$7x$ is an algebraic term in which 7 is a _____ and x is a _____.

Answer (Please select your correct option)



term, expression



coefficient, variable

Correct Answer Solved By Hadi
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03228043306



variable, coefficient



numerical, alphabet

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$9x^2 + 3x + 4$ is _____.

Answer (Please select your correct option)



an equation



a term



an algebraic expression



quadratic equation

Correct Answer Solved By Hadi
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$$x_1 - 2x_2 + x_3 = 0$$

Which of the following is the coefficient matrix for the system $2x_2 - 7x_3 = 8$?

$$-4x_1 + 3x_2 + 9x_3 = -6$$

Answer (Please select your correct option)

☐ $\begin{bmatrix} 1 & -2 & 1 \\ 0 & 2 & -7 \\ 4 & 3 & 9 \end{bmatrix}$

Correct Answer Solved By Hadi
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☐ $\begin{bmatrix} 1 & -2 & 0 \\ 0 & 2 & 8 \\ 4 & 3 & 9 \end{bmatrix}$

☐ $\begin{bmatrix} 1 & 1 & 0 \\ 0 & -7 & 8 \\ 4 & 3 & 9 \end{bmatrix}$

☐ $\begin{bmatrix} 1 & 0 & -4 \\ -2 & 2 & 3 \\ 4 & 3 & 9 \end{bmatrix}$

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Two simultaneous linear equations in two variables have no solution if their corresponding lines are _____.

Answer (Please select your correct option)

☐ parallel and distinct

Correct Answer Solved By Hadi
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☐ intersecting

☐ coincident

☐ perpendicular

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If reduced echelon form of a linear system is $\begin{bmatrix} 1 & 0 & 5 & 5 \\ 0 & 1 & 1 & 6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ when free variable $x_3 = 0$, then which of the following is true for it?

Answer (Please select your correct option)

☐ The particular solution is $(0, 5, 6)$.

☐ The particular solution is $(6, 5, 0)$.

☐ The particular solution is $(5, 6, 0)$.

☐ The particular solution is $(0, 6, 5)$.

Correct Answer Solved By Hadi
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If A is an $m \times n$ matrix and $Ax = b$ has a solution then which of the following is true?

Answer (Please select your correct option)

☐

A has a pivot position in every row.

Correct Answer Solved By Hadi
usmanraj20@gmail.com
03228043306

☐

$\det(A) = 0$.

☐

Echelon form of A has at least one row of zeros.

☐

$b \notin \text{Span}(a_1, a_2, \dots, a_n)$.



If a homogeneous system $Ax = 0$ has a trivial solution, then which of the following is (are) the value(s) of the vector x ?

Answer (Please select your correct option)

☐ -1

☐ 0

☐ 1

☐ 2

Correct Answer Solved By Hadi
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If $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation defined by $T(\vec{x}) = A\vec{x}$ (for all $\vec{x} \in \mathbb{R}^n$), then which of the following is true for A ?

Answer (Please select your correct option)



A is a singular matrix.



A is a square matrix.



A is a unique matrix.

Correct Answer Solved By Hadi
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A is not a unique matrix.

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Which of the following is true for the linear operator L defined by $L \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} a_1 \\ -a_2 \end{pmatrix}$?

Answer (Please select your correct option)

☐ It is an enlargement by a negative scale factor.

☐ It is a shear.

☐ It is a reflection about X -axis.

☐ It is a reflection about Y -axis.

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If T be a transformation, then which of the following is true for its linearity ?

Answer (Please select your correct option)

☐

$T(c\hat{u} + d\hat{v}) = c T(\hat{u}) + d T(\hat{v})$; where 'c' and 'd' are scalars

☐

$T(c\hat{u} + d\hat{v}) = c T(\hat{u}) + d T(\hat{v})$; where 'c' and 'd' are scalars

☐

$T(c\hat{u} \times d\hat{v}) = c T(\hat{u}) \times d T(\hat{v})$; where 'c' and 'd' are scalars

☐

$T(c\hat{u} + d\hat{v}) = d T(\hat{u}) + c T(\hat{v})$; where 'c' and 'd' are scalars

Correct Answer Solved By Hadi
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If $A = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1+1 & 2-1 \\ 2+2 & 4-1 \end{bmatrix}$, then which of the following is true for A and B ?

Answer (Please select your correct option)

☐ A and B are equal matrices.

Correct Answer Solved By Hadi
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03228043306

☐ A is the transpose of B .

☐ B is the transpose of A .

☐ B is the multiplicative inverse of A .



What is the maximum possible number of pivots in a 6×6 matrix ?

Answer (Please select your correct option)

☐ 2☐ 4☐ 6☐ 0

Correct Answer Solved By Hadi
usmanraj20@gmail.com
03228043306

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If $A = \begin{bmatrix} P \\ Q \end{bmatrix}$ and $B = \begin{bmatrix} R & S \end{bmatrix}$ (where P , Q , R and S are square sub-matrices of same size), then which of the following is possible?

Answer (Please select your correct option)

☐

The product AB is defined but BA is not defined.

☐

The product AB is not defined but BA is defined.

Correct Answer Solved By Hadi
usmanraj20@gmail.com
03228043306

☐

The product AB and BA both are not defined.

☐

The product AB and BA both are defined.

Which of the following is true for the matrix $A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ O & A_{22} & A_{23} \\ O & O & A_{33} \end{bmatrix}$; where A_{11} , A_{22} and A_{33} are square sub-matrices, and 'O' is a zero sub-matrix?

Answer (Please select your correct option)

☐ It is a block upper triangular matrix.

Correct Answer Solved By Hadi
usmanraj20@gmail.com
03228043306

☐ It is a block lower triangular matrix.

☐ It is diagonal-constant matrix.

☐ It is a Null matrix.



If a matrix A is factorized into lower and upper triangular matrices , then which of the following is true for the matrix ?

Answer (Please select your correct option)

☐ It is called an LU -procedure.

☐ It is called an LU -decomposition.

☐ It is called an LU -matrices.

☐ It is called an LU -algorithm.

Correct Answer Solved By Hadi
usmanraj20@gmail.com
03228043306

If $A = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$, then which of the following is the minor of entry a_{21} ?

Answer (Please select your correct option)

☐ 2

☐ 3

☐ 4

☐ 5

Correct Answer Solved By Hadi
usmanraj20@gmail.com
03228043306



Which of the following is true for the coefficient matrix in the Cramer's Rule ?

Answer (Please select your correct option)

☐

It must be invertible .

Correct Answer Solved By Hadi
usmanraj20@gmail.com
03228043306

☐

It must be singular .

☐

It may or may not be invertible .

☐

It may or may not be non - singular .

Made By: Waqar Siddhu



If a set W be a subspace of a vector space V , then which of the following is NOT true for it ?

Answer (Please select your correct option)



It must be closed under the scalar multiplication .



It may or may not be closed under the operation of addition .



It must have an additive inverse of each element .



It must be commutative under the operation of addition .

Correct Answer Solved By Hadi
usmanraj20@gmail.com
03228043306



If the set $V = \{0, 1, 2, 3\}$, then which of the following is true for it ?

option 3 or 4

Answer (Please select your correct option)



It is closed under operation of addition .



It is a vector space .



It is commutative under operation of addition .



It has an additive inverse of each element .

Made By: Waqar Siddhu

If $c_1 \hat{v}_1 + c_2 \hat{v}_2 + c_3 \hat{v}_3 + \dots + c_p \hat{v}_p = 0$, and vectors $\hat{v}_1, \hat{v}_2, \hat{v}_3, \dots, \hat{v}_p$ all are linearly independent then which of the following is true ?

if dependent
then option 2

Answer (Please select your correct option)

☐ $c_1 = c_2 = c_3 = \dots = c_p = 0$

Correct Answer Solved By Hadi
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03228043306

☐ $c_1 \neq c_2 \neq c_3 \neq \dots \neq c_p \neq 0$

☐ $c_1 \neq c_2 = c_3 = \dots = c_p \neq 0$

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$7x$ is an algebraic term in which 7 is a _____ and x is a _____.

Answer (Please select your correct option)

☐

term, expression

☐

coefficient, variable

Correct Answer Solved By Hadi
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☐

variable, coefficient

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$9x^2 + 3x + 4$ is _____.

Answer (Please select your correct option)



an equation



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☐
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☐ 6

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Answer (Please select your correct option)



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It may or may not be closed under the operation of addition .



It must have an additive inverse of each element .



It must be commutative under the operation of addition .

Correct Answer Solved By Hadi
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03228043306



If the set $V = \{0, 1, 2, 3\}$, then which of the following is true for it ?

option 3 or 4

Answer (Please select your correct option)



It is closed under operation of addition .



It is a vector space .



It is commutative under operation of addition .



It has an additive inverse of each element .

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then option 2

Answer (Please select your correct option)

☐ $c_1 = c_2 = c_3 = \dots = c_p = 0$

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☐ $c_1 \neq c_2 \neq c_3 \neq \dots \neq c_p \neq 0$

☐ $c_1 \neq c_2 = c_3 = \dots = c_p \neq 0$

☐ $c_1 \neq c_2 = c_3 = \dots = c_p = 0$



$7x$ is an algebraic term in which 7 is a _____ and x is a _____.

Answer (Please select your correct option)



term, expression



coefficient, variable



variable, coefficient



numerical, alphabet

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Which of the following is the pivot element in the second row of the matrix?

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 8 \\ 0 & 0 & -3 & 5 & 1 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & -3 \end{pmatrix}$$

Answer (Please select your correct option)

0



1



-3



5



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Which of the following is true for the matrix $\begin{pmatrix} 1 & 3 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$?

Answer (Please select your correct option)

☐ It is an identity matrix.

☐ It is in reduced echelon form.

☐ It is in echelon form.

☐ It is a rectangular matrix.

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If reduced echelon form of a linear system is $\begin{bmatrix} 1 & 0 & 5 & 5 \\ 0 & 1 & 1 & 6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ when free variable $x_3 = 0$, then which of the following is true for it?

Answer (Please select your correct option)

☐ The particular solution is $(0, 5, 6)$.

☐ The particular solution is $(6, 5, 0)$.

☐ The particular solution is $(5, 6, 0)$.

☐ The particular solution is $(0, 6, 5)$.

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If $\vec{b} = c_1 \vec{u}_1 + c_2 \vec{u}_2$ and $\vec{a} = d_1 \vec{u}_1 + d_2 \vec{u}_2$ where c_1, c_2, d_1 and d_2 are scalars, then which of the following options is correct?

Answer (Please select your correct option)

☐ Only $\vec{b} \in \text{Span}\{\vec{u}_1, \vec{u}_2\}$.

☐ Only $\vec{a} \in \text{Span}\{\vec{u}_1, \vec{u}_2\}$.

☐ Both $\vec{a}, \vec{b} \in \text{Span}\{\vec{u}_1, \vec{u}_2\}$.

☐ Both $\vec{a}, \vec{b} \notin \text{Span}\{\vec{u}_1, \vec{u}_2\}$.

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If a homogeneous system $Ax = 0$ has a trivial solution, then which of the following is (are) the value(s) of the vector x ?

Answer (Please select your correct option)



-1



0



1



2

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If a homogeneous system $Ax = 0$ has non-trivial solution, then which of the following is true for the system?

Answer (Please select your correct option)



The system has at least no free variable.



The system has at least one free variable.

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The system has at least two free variables.



The system has at least three free variables.

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If $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1+6x_3 \\ 3 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 6 \\ 0 \\ 1 \end{bmatrix}$ be the general solution of $Ax = b$, then which of the following is parametric equation for the given solution?

Answer (Please select your correct option)

☐

$$p = s x + t u + v$$

☐

$$x = p + t v$$

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☐

$$v = p + t x$$

☐

$$t x = p + v$$

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If the equation $T(\vec{x}) = \vec{0}$ has only the trivial solution, then which of the following is true for the linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$?

Answer (Please select your correct option)



T is one-to-one.

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T is onto.



T is a rotation.



T is a reflection.

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If linear transformation $T : R^n \rightarrow R^m$ is transformed into a matrix A , then which of the following is the order of A ?

Answer (Please select your correct option)

☐ $n \times m$

☐ $m \times m$

☐ $m \times n$

☐ $n \times n$

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If $A = A^t$ (where A is a square matrix) , then which of the following is the most appropriate option for A ?

Answer (Please select your correct option)



A is an invertible matrix.



A is a singular matrix.



A is a symmetric matrix.

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A is a scalar matrix.

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If A is an invertible matrix , then which of the following is true ?

Answer (Please select your correct option)

☐ $(A^{-1})^{-1} = A$

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☐ $(A^{-1})^{-1} = \frac{1}{A}$

☐ $(A^{-1})^{-1} = A^{-1}$

☐ $(A^{-1})^{-1} = \det(A)$



If $A = \begin{bmatrix} 4 & -1 \\ 5 & 3 \end{bmatrix}$, then which of the following is the value of $\det(A)$?

Answer (Please select your correct option)

☐

7

☐

-17

☐

17

☐

11

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If a system of equations is solved using the Jacobi's method , then which of the following is NOT true about the matrix M that is derived from the coefficient matrix ?

Answer (Please select your correct option)



All of its entries below the diagonal must be zero .



All of its entries above the diagonal must be zero .



It may or may not be invertible .



It is a non-singular matrix .

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If the matrix $A = \begin{bmatrix} 5 & 4 \\ 3 & 3 \end{bmatrix}$, then which of the following is the most suitable option for it ?

Answer (Please select your correct option)

☐

It is not strictly diagonally dominant.

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☐

It is strictly diagonally dominant.

☐

It is not diagonally dominant.

☐

It is diagonally dominant.

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If $A = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$, then which of the following is the minor of entry a_{21} ?

Answer (Please select your correct option)

2



3



4



5



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If $A = \begin{bmatrix} 4 & 5 \\ 6 & 7 \end{bmatrix}$, then which of the following is the value of C_{21} ?

Answer (Please select your correct option)



5



-5



6



-6

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If the determinant of the matrix $A = \begin{bmatrix} 2 & 3 & 4 \\ 1 & 2 & 2 \\ 3 & 4 & 5 \end{bmatrix}$ is -1 and the matrix B is obtained by adding 2 times of the second row in the first row of the matrix A , then which of the following is true about the matrix B ?

Answer (Please select your correct option)

☐ Its determinant is -1 .

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☐ Its determinant is 1 .

☐ Its determinant can not be evaluated .

☐ The information is not sufficient to calculate the determinant .



If the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix}$, then which of the following is true about it ?

Answer (Please select your correct option)



Its determinant is 0 .



Its determinant is 1 .



Its determinant is 2 .

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Its determinant is 4 .

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If a set $S = \{ v_1 = (1, 2), v_2 = (4, 8) \}$, then which of the following is the most appropriate option ?

Answer (Please select your correct option)



It is a basis of R^2 .



It is linearly independent.



It spans R^2 .



It is linearly dependent.

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